MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 12 (8-12 Nov 2026) - Period 1	Unit 4 - 4.1 Setting the Game Elements (2/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 12 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 70% ordered listener correctly; sessionStorage understood by 80%. Today: loops and asynchronous timing for the game.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use for/while loops and asynchronous methods (setTimeout, setInterval, clearInterval) to repeat actions and time events. Sprint 4.1 obj. 3-4; Challenges 2-3; CSTA 3A-AP-15.

Success criteria — *what will successful learning look / sound like?*

I write a for loop to repeat code; I use setTimeout/setInterval with milliseconds correctly; I stop a timer with clearInterval; my countdown works.

Key knowledge / skill / understanding:

Loops repeat; 1 s = 1000 ms; asynchronous code runs later, not in order.

Likely misconception / access or challenge need:

setTimeout pauses the program; loops always need a counter from 0.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*	ACTIVITY 2 *student task + evidence of learning*	ACTIVITY 3 *student task + evidence of learning*
Strategy: Inquiry / investigation **Time:** 8 min	**Strategy:** Problem solving **Time:** 20 min	**Strategy:** Cooperative learning **Time:** 8 min
What students will do: Predict-then-run: 4 short code snippets (loop, setTimeout order). Students predict console output, run, then explain differences.	**What students will do:** Challenge 2: countdown timer from 10 that disappears at 0 and shows a birthday message (or Eid/National Day greeting); then add a loop creating 9 holes dynamically.	**What students will do:** Pairs swap code and trace each other's timer with a trace table (time, value, output).
Evidence tool: Mini-whiteboards Assessment: ☑ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☑ Pair ☐ Group	**Evidence tool:** Practical / performance task Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☑ Individual ☐ Pair ☐ Group	**Evidence tool:** Peer-assessment checklist Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☐ Individual ☑ Pair ☐ Group
Resources: Snippets in browser console.	**Resources:** VS Code; QR 'Asynchronous methods'; source files.	**Resources:** Trace table template.
Support (low performers / SEN): Annotated snippets; buddy support.	**Support (low performers / SEN):** Countdown starter with comments; flowchart.	**Support (low performers / SEN):** Partially filled trace table.
Stretch (fast learners / gifted): Explain why setTimeout(...,0) still runs last.	**Stretch (fast learners / gifted):** Challenge 3: print primes up to n using loops and if-else.	**Stretch (fast learners / gifted):** Add an edge case (n = 0 or negative) to test.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 7 min

Hook: a countdown on the board - how do games keep time? Model a for loop with a trace table; model setInterval with a counter and clearInterval; convert seconds to ms on the board. Share success criteria.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students choose the celebration message (birthday, Eid, Diwali, National Day) - culturally responsive context; trace tables build computational thinking.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Infinite loops freezing the tab; forgetting clearInterval; using seconds not ms; counter declared inside the interval. Response: 'my timer goes negative'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: write a setInterval countdown from 5 in pseudocode; convert 2.5 s to ms. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: finish countdown; try the primes challenge (optional). Review student book 4.2 'Start the game'.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Predict-then-run created real surprise and talk. Celebration choice made it personal and inclusive. 80% built a working countdown.

What will I change next time?

Show how to recover from an infinite loop (close tab) before practice. Trace tables need a worked example first.

Students needing follow-up:

Students who froze browsers repeatedly (10/4, 10/5) need loop-condition practice. Inform parents of 5 students behind on submissions.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☐3 ☑C ☐H	☑1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback